

ISGEC Internship Progress Log

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Period: 8 June 2026 – 16 July 2026

Date	Task	Description
8–10 Jun	Machine Familiarisation	In-depth study of SCR machine — operation, component breakdown, load paths, GA drawing and CAD model review
11 Jun AM	FEMAP Exercise	1D bar axial load analysis in FEMAP/Nastran — software familiarisation and result verification
11 Jun PM	FEA Theory	Structural analysis theory study — FEM Section II, load classifications, load combinations, etc.
12 Jun AM	FEA Theory	FEM Section II continued, further literature on FEA & structural analysis reviewed
12 Jun PM	Load Data	Excel sheet preparation commenced — non-structural equipment weights and mounting point locations
15 Jun	Load Data / Mid Surface	Excel sheet completed; mid surface creation commenced — rear boom section
16–20 Jun	Mid Surface	Rear and mid boom sections mid surface creation completed in SolidWorks
23–25 Jun	Mid Surface	Front boom section mid surface creation completed — full boom mid surface model complete
26 Jun	Colour Coding	Colour coding of boom mid surfaces by plate thickness commenced
29 Jun – 3 Jul	Boom Structure — Assembly & Colour Coding	Assembly of rear, mid, and front boom sections; minor corrections made to mid surface geometry; colour coding of boom structure completed
6–9 Jul	Mid Surface — Mast/Pylon	Mast/pylon mid surface file reviewed and found significantly incomplete — numerous gaps and missed thickness offsets identified. Mid surface completion work carried out to close gaps and finish offset alignment across the full mast structure
10 Jul	Colour Coding — Counterweight Boom & Mast	Colour coding of counterweight boom and mast/pylon mid surfaces by plate thickness commenced
13 Jul	Colour Coding — Counterweight Boom & Mast (contd.)	Mid surface finishing and colour coding of counterweight boom and mast/pylon completed
14–15 Jul	Assembly — Full SCR Structure	Boom structure, mast/pylon, and counterweight boom mid surface models assembled into the complete SCR upper structure. During assembly, overlapping plates were identified at the boom rear-to-front junction that had not been separately colour coded following the earlier rear/mid/front assembly step — these were corrected. Full mid surface model and colour coding of the complete SCR structure completed by end of 15 July
15 Jul (PM) – 16 Jul (AM)	Mid Surface Analysis & Quality Check (QC)	Wireframe review of the assembled mid surface model to check for gaps, free edges, and geometric errors; detailed minute-level analysis of the complete SCR mid surface structure conducted
16 Jul (PM)	FEMAP Preparation	FEMAP training exercises undertaken from “FEMAP 101 for FEMAP v12 — Student Workbook” (Siemens PLM Software, 2019); continued study of applicable ISO standards and FEM theory in preparation for analysis phase